SIMULATION OF PV/WIND/ DIESEL HYBRID POWER SYSTEMS FOR RURAL ELECTRIFICATION IN ALGERIA

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ABSTRACT**:** Adrar region is a typical rural area in Algeria where connection to the national grid is not possible even in the future. Diesel is the main fuel for full filling the energy demand. Solar and Wind resources are the hybrid options for this site. The HOMER energy modeling software is available for optimization of renewable based energy efficient systems for single and up to 20 homes. It shows that the energy costs are more attractive for the electrification of 20 homes where per unit (kWh) cost of energy varies from 2.23 to 1.759 USD. For a single –wire grid extension cost of 8000$/km, operation and maintenance cost of 160 $/yr./km and grid power price of 0.1 $/km, the breakeven grid extension distances were found to be 3.71 km for single home and 53.1km for 20 homes respectively. On the other hand, it is found that the optimum results of hybrid renewable system show a 91 % and 92 % reduction of emissions respectively for a single home as well as for 20 homes. The RF of the optimized system evaluates to, respectively, 91 % and 92 % for single and 20 homes relative to the studied location.
Keywords: Hybrid system, rural electrification, cost of energy, breakeven grid distance, emission.

1 INTRODUCTION

In remote Algerian villages, far from the grids, electric energy is usually supplied by diesel generators. In most of these cases, the supply with diesel fuel becomes highly expensive while hybrid/photovoltaic/wind generation becomes competitive with diesel only generation [1], [2]. Photovoltaic/wind/diesel hybrid systems are more reliable in producing electricity than photovoltaic-only/wind-only systems, and often present the best solution for remote areas electrification. The diesel generator reduces the photovoltaic/wind component count while the photovoltaic/wind systems decrease the operating time duration of this generator, reducing the running costs of the system [3]. On the other hand, the addition of storage reduces the start/stop cycles of the generator, and thus, considerably reduces the fuel consumption [4], [5].

In order to effectively explain the benefits of Hybrid system a software package called HOMER [6] was used in this study. HOMER is sophisticated software developed by the National Renewable Energy (RE) Laboratory for analyzing the economics of small power systems. After inputting the solar/wind resources along with the cost of equipment, HOMER crunches the numbers to give the “Optimizing System Type” based purely on economics and availability of resources.

Analysis has been done for the single home user as well as for a combination of 20 home users to get the most economic, technical, breakeven grid distance, emissions and environment viable options. For the purpose of illustration, one Algerian rural site has been considered (Adrar).

Many hybrid power systems have been proposed in the past for remote areas but the vast majority of them were based on the technical PV-diesel or PV-Wind systems [7-19]. Many tools are also available for sizing and simulation of hybrid power systems but few of them include the technical and economic study of the PV-Wind-Diesel hybrid system with battery storage using different load combinations. The authors have adopted a new approach to the PV-Wind-Diesel hybrid system with battery storage including the technical, economic, breakeven grid distance and environmental study using two load combinations for a site selected from an Algerian rural area.

2 USED DATA

In the present study, an Algerian site has been selected for which solar radiation; ambient temperature and wind speed were available. Thus, the geographical coordinates and the climate type classification according to Koppen Geiger [20] were also given, see Table 1.

Table I the geographical coordinates and the climate type classification

Site	Latitude (°)	Longitude (°)	Altitude (m)	Climate type
Adrar	27.8 North	0.18 West	264	Bwh[20]*

**BWh* is a dry arid *climate* found in low latitude deserts

3 PRESENTATION OF THE CASE STUDY

A hybrid energy system generally consists of primary energy sources working in parallel with standby secondary energy storage units. HOMER has been used to optimize the system for the best system energetic efficiency for Adrar region considering different load and Wind-PV-Diesel combinations based on the relative solar/wind resources. Fig.1 shows the schematic diagram representing the scheme as implemented in the HOMER simulation tool. HOMER simulates the operation of the considered system by making energy balance calculations for each of the 8760 hours in a year.

For each hour, HOMER compares the electric and thermal demand in the hour to the energy that the system can supply in that hour and calculates the flows of energy to and from each component of the system. For systems that include batteries or fuel-powered generators, HOMER also decides for each hour how to operate the generators and whether to charge or discharge the batteries.

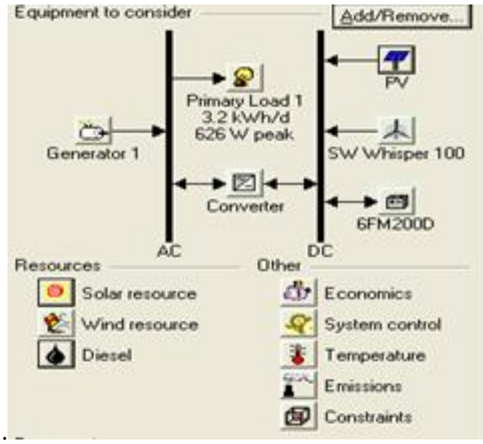


Fig. 1 Hybrid energy system with HOMER

HOMER performs these energy balance calculations for each system configuration under consideration. It then determines whether a configuration is feasible, i.e., whether it can meet the electrical demand under the specified conditions and estimates the cost of installing and operating the system over the lifetime of the project. The system cost calculations account for costs such as capital, replacement, operation and maintenance, fuel and interest. Information about the load, resources, economic, constraints, controls and other components used in HOMER for this case are given below:

3.1 Performance metrics

In this study, total NPC, COE and RF have been considered as performance metrics to evaluate and compare different systems. The total NPC of a system is the present value of all the costs that it incurs over its lifetime, minus the present value of all the revenues that it earns over its lifetime. Costs include capital costs, replacement costs, operational and maintenance costs, fuel costs, and emissions penalties.

On the other hand, COE is the average cost per kWh of electricity. To calculate the COE, HOMER divides the annualized cost of producing electricity (the total annualized cost minus the cost of serving the thermal load) by the total useful electric energy production. The NPC is a more trustworthy number than the COE, therefore in this analysis, NPC has been taken as the primary metric.

The RF is that portion of the system's total energy production originating from renewable power sources. HOMER calculates the RF by dividing the total annual renewable energy production by the total energy production.

3.2 Electric load

A typical load system (Table 2) for a single home in remote areas has been considered for the analysis. Monthly average hourly load demand has been given as an input to HOMER which then generates the daily and monthly load profile for a year for 20 homes~~.~~ (see Fig. 2). It is found that for this system each home user consumes around 3200Wh/day with a peak demand of nearly 626W.

Table II Appliances for single home user [2]

		Power (W)	*Duration of consumption (hours)*	Day load (wh)
Lighting	Rooms	22	8	176
	Lounge	22	6	132
	Corridor	22	4	88
	Bathroom	22	4	66
	Toilet	22	1	22
	Kitchen	11	7	77
Equipments	Refrigerator	120	12	1440
	TV	80	7	560
	FAN	100	4	400
	Various	100	2	200
	Total load			3198

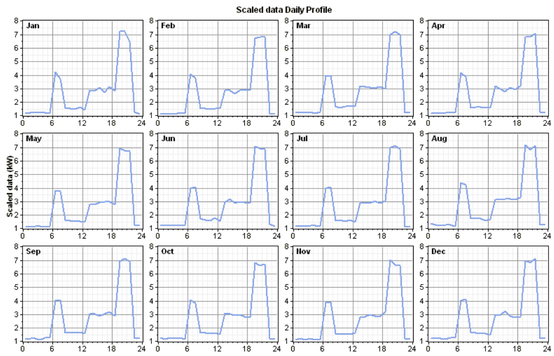


Fig.2. Monthly average hourly load profile for 20 home-users

3.3 Renewable resources

The hourly global radiation data have been taken from (WMO/OMM) [21] while HOMER introduces clearness index from the latitude information of the selected site (Fig. 3): (a) hourly global radiation data and (b) hourly ambient temperature data.

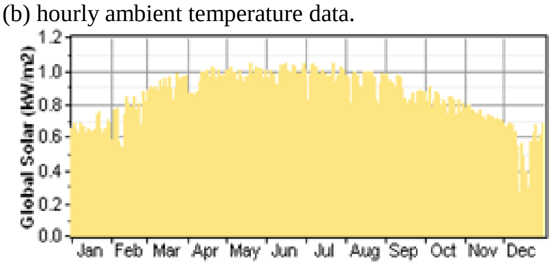


Fig.3 (a) Hourly global radiation Data

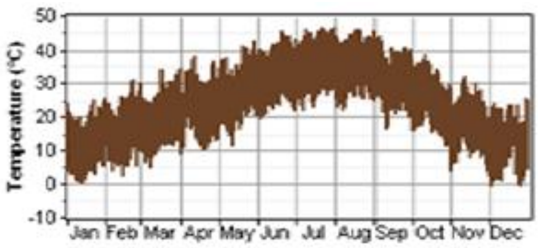


Fig.3 (b) Hourly ambient temperature Data

For wind component, hourly measured data were used considering a height of 10m (WMO/ OMM) [21]. HOMER then generated and synthesized the monthly average data based on the other parameters such as:

Weibull factor k = 2.35, Autocorrelation factor 0.961, Diurnal pattern strength (wind speed variation over a day) = 0.153, Hour of peak wind speed = 16 to generate hourly data for a year.

Fig. 4 shows (a) the wind speed probability distribution function and (b) averaged hourly wind speed for 1 year with the annual average wind speed equal to 7.81 m/s.

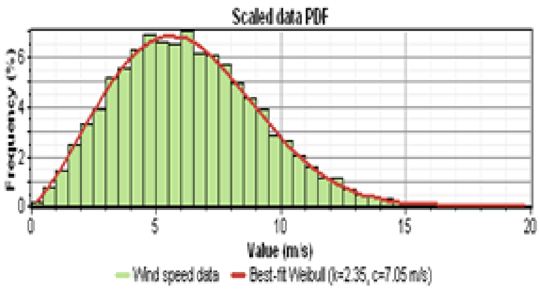


Fig. 4 (a) Wind speed probability density function

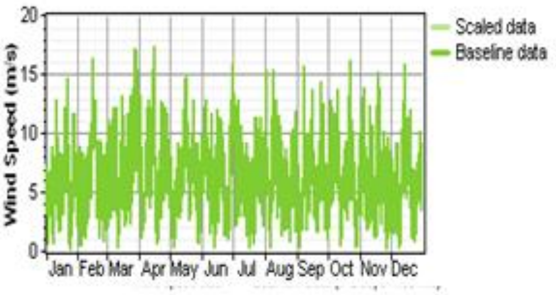


Fig. 4 (b) Daily wind speed

4 HYBRID SYSTEM COMPONENTS

4.1 Photovoltaic module

The cost of PV module including installation has been considered as 4 USD/W for Algeria (1USD=78 DZD, 2013 [22]). The lifetime of the modules has been taken as being 25 years and these are titled like the location latitude with no tracking mode.

4.2 Wind generator

The load demand is very low for a single-home system and the price per kW turbine cost is very high for low capacity wind turbines compared to that of high capacity ones. Also low capacity wind turbines are not much available. Now–a–days, research and development are going on to improve the technology and design of low capacity turbines with low cut-in speed around 2.5 m/s. So, for the Adrar site which presents a high wind potential, a turbine from Southwest Wind Power (model Whisper 100; capacity 0.9 kW) has been considered at the cost of 10910 USD including tower and installation costs.

4.3 Diesel generator

Diesel generator supplies a significant amount of energy for up to 20 houses. The diesel generators selected operation time duration, minimum load ratio and power, which is generated by a fuel consumption of 0.25 l / hr, are respectively 15000, 30% and 0.2 kW to 5 kW.

4.4 Battery with controller

As the system considered the AC load only, battery and controller were also as main part of the system. Battery from Vision Battery Company (Model Vision 6FM200D; nominal voltage: 12V; nominal capacity: 200Ah) has been used at a cost of 853 $, the battery being supplied with charge controller.

4.5 Economics and constraints

The project life time has been considered to be 25 years and the annual interest rate has been taken as being 10%. As the system has been designed for up to 20-house users but at low user's load consumption, operation and maintenance costs have been taken as being 300 $/year considering the system being free from capacity shortages and the operating reserve is 10% of hourly load.

5 OPTIMIZATION AND SENSITIVITY ANALYSIS

To evaluate the performance of different hybrid systems in this study, optimal systems and sensitivity analysis have been performed using HOMER simulation tools. Sensitivity analysis and optimization results have given a detailed scenario for either single-home or 20-home–case: either it is or it is not economically or environmentally feasible in specific meteorological conditions. Sensitivity analysis is a measure that checks the sensitivity of a model and also changes of the structure of the model [6, 23].

This analysis is useful for helping with decision making or the development of recommendations for the model. In this paper, sensitivity analysis has been performed to study the effects of variation in the wind speed and solar irradiation and fuel price, and to make appropriate recommendations for developing a hybrid RE system. The simulation software simulates the long term implementation of the hybrid system based on respective size searches for predefined sensitivity values of the components. Different NPC, COE and RF values have been used as model input. HOMER simulation is then performed for every possible system combination for a period of one year.

6 EMISSION ANALYSIS

The use of RE ensures a decrease~~s~~ in the emission of CO_2, SO_2 and NOx into the atmosphere, which helps protect the environment from global warming. This study estimates the total yearly emissions by standard Diesel, Wind/Diesel, PV/Diesel and PV/Wind/Diesel system with battery storage.

7 RESULTS AND DISCUSSION

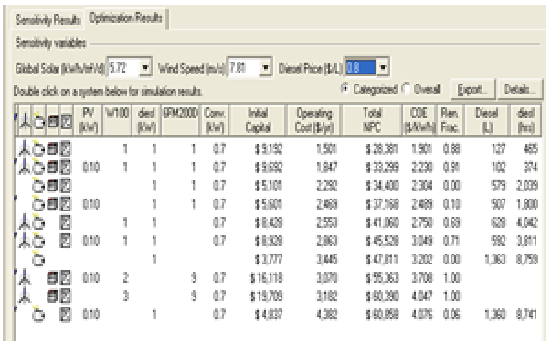
In this section, the results obtained using HOMER software for an Algerian rural site and for a system designed for up to 20-house users are presented.

7.1 Optimization results

Models were developed for the selected site in Algeria using HOMER simulation tools. The optimization results for this location for a specific wind speed (7.25 m/s), solar radiation (5.72 kWh/m²), and Diesel price ($0.8/l) are illustrated in Fig. 5 (a) for a single-house and Fig. 5 (b) for 20-house users.

Diesel, Wind/Battery/Diesel, PV/Battery/Diesel and PV/Wind/Battery/Diesel systems, for single and also for 20-house users, were studied using HOMER simulation software and their performances compared based on performance metrics and locations.

The performance metrics NPC, COE, RF and GHG emission were measured with optimization modeling and sensibility analysis undertaken to analyze the performances of the PV/Wind/Battery/Diesel hybrid systems for single and also for 20-house users. HOMER simulates all the systems in their respective predefined search spaces, and simulations are undertaken for every possible system combination and configuration for a period of one year. The sensitivity variables have been set for solar irradiation, wind speed and Diesel price. A total of 180 sensitivity cases were run for each system configuration. From the results, it can be seen that the RF is respectively 91 %, 92 % for single home and 20 homes relative to the studied location.


Sensitivity Results | Optimization Results

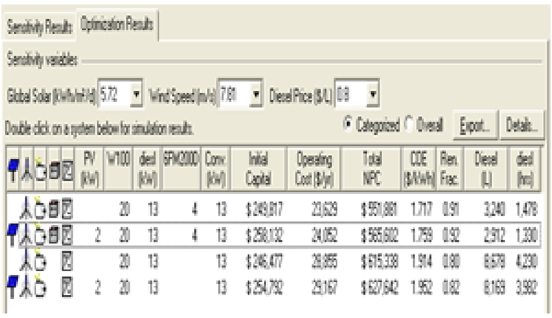
Sensitivity variables

Global Solar (kWh/m²/d) 5.72 Wind Speed (m/s) 7.81 Diesel Price ($/L) 0.8

Double click on a system below for simulation results. Categorized Overall Export... Details...

PV (kW)	W100	died (kW)	6FM200D	Conv. (kW)	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.	Diesel (L)	died (hrs)
	1	1	1	0.7	$ 9,192	1,501	$ 28,381	1.901	0.88	127	485
0.10	1	1	1	0.7	$ 9,692	1,847	$ 33,299	2.230	0.91	102	374
		1	1	0.7	$ 5,101	2,292	$ 34,400	2.304	0.00	579	2,009
0.10		1	1	0.7	$ 5,601	2,469	$ 37,168	2.489	0.10	507	1,800
	1	1		0.7	$ 8,429	2,553	$ 41,060	2.750	0.69	629	4,042
0.10	1	1		0.7	$ 8,929	2,863	$ 45,528	3.049	0.71	592	3,811
		1			$ 3,777	3,445	$ 47,811	3.202	0.00	1,363	8,759
0.10	2		9	0.7	$ 16,118	3,070	$ 55,363	3.708	1.00		
	3		9	0.7	$ 19,709	3,182	$ 60,390	4.047	1.00		
0.10		1		0.7	$ 4,837	4,382	$ 60,858	4.076	0.06	1,360	8,741

Fig.5 (a) Optimization results of ADRAR for a single home with 5.72kWh/m²/day solar radiation, and 7.81m/s wind speed


Sensitivity Results | Optimization Results

Sensitivity variables

Global Solar (kWh/m²/d) 5.72 Wind Speed (m/s) 7.81 Diesel Price ($/L) 0.8

Double click on a system below for simulation results. Categorized Overall Export... Details...

PV (kW)	W100	died (kW)	6FM200D	Conv. (kW)	Initial Capital	Operating Cost ($/yr)	Total NPC	COE ($/kWh)	Ren. Frac.	Diesel (L)	died (hrs)
	20	13	4	13	$ 249,817	23,629	$ 551,881	1.717	0.91	3,240	1,478
2	20	13	4	13	$ 258,132	24,052	$ 565,602	1.759	0.92	2,912	1,330
	20	13		13	$ 246,477	28,855	$ 615,338	1.914	0.80	8,678	4,230
2	20	13		13	$ 254,792	29,167	$ 627,642	1.952	0.82	8,169	3,982

Fig.5 (b) Optimization results of ADRAR for 20 homes with 5.72kWh/m²/day solar radiation, and 7.81m/s wind speed

The monthly average electric energy production in Adrar area is represented in Fig. 6 (a) for single home and in Fig. 6 (b) for 20 homes. From the figures, it is seen that the wind energy has a significant contribution in the hybrid model, and contributed 86% and 88% respectively for single home and 20 homes. On the other hand, the PV array contributed only 5%, 4 % and the Diesel contributed the remaining 9 %, 8% of the total energy production respectively for single home and 20 homes.

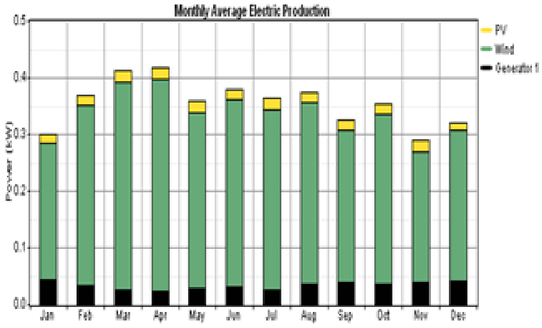


Fig. 6(a) Monthly average PV/Wind /Diesel electric production of Adrar, Algeria for a single home

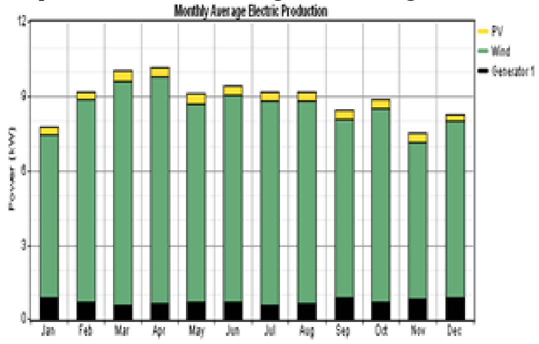


Fig. 6(b) Monthly average PV/Wind /Diesel electric production of Adrar, Algeria for 20 homes

The optimization results related to Adrar region are illustrated in Fig. 5 (a) and Fig. 5 (b) respectively for single home and 20 homes. It is seen that wind/Battery/Diesel power systems are economically more feasible with a minimum total NPC of $28,381 and $551,881 and a minimum COE of $1.901 and $1.717 respectively for single home and 20 homes than the PV/Wind/ Battery/Diesel hybrid systems with a minimum total NPC of $32,299 and $565,602 and a minimum COE of $2.23 and $1.759 respectively for single home and 20 homes.

Fig. 7 (a) and Fig.7 (b) shows the cash flow summary for Adrar zone, south west territory respectively for single home and 20 homes. From the figures, it can be seen that most of the capital cost is required for the wind turbine and Diesel generator, and they equally share 40% of the total capital cost for single home and 20 homes. It also can be seen that most of the capital cost is related to the wind turbine, and it corresponds to 89% of the total capital cost. However, most of the operating cost is due to the wind turbine, PV components, and converters for single home and is due to the wind turbine and converter for 20 homes.

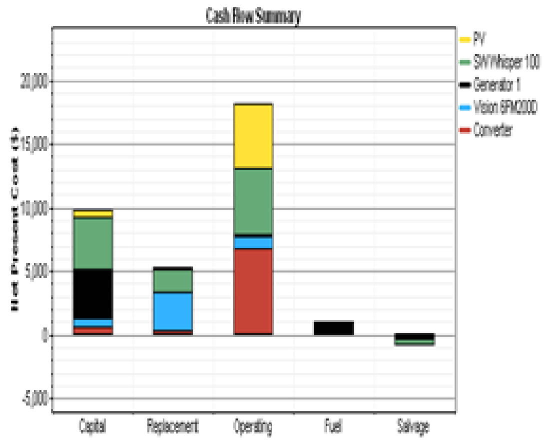


Fig.7(a) Cash flow summary for Adrar single home

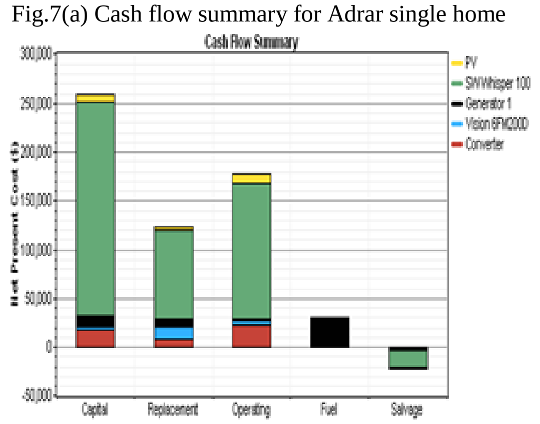


Fig.7(a) Cash flow summary for Adrar 20 homes

7.2 Comparison between single home and 20 homes cost

Analysis shows that the COE (kWh) is low for the system which is the combination of 20 homes. On the other hand, it also reveals that the energy cost depends largely on the RE potential quality. Table 3 shows the load demand for each combination of homes and each site together with system architecture and a financial summary.

7.3 Sensitivity Results

From the sensitivity analysis, the models were analyzed to explore their characteristics using optimal system type (OST). Fig. 8 shows the sensitivity results from the data collected from ADRAR. It can be seen that a Wind/Diesel/Battery system is only suitable when the wind speed is above 5.4 m/s.

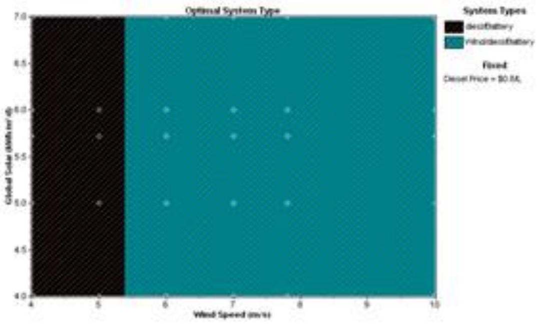


Fig. 8 Sensitivity results with fuel prices are fixed \$0.8/l

Table III HOMER analysis and results

Site	PV module (kW)	Wind Generator (Quantity)	Diesel Generator (kW)	Battery (Quatity)	Initial cost ($)	Total NPC ($)	COE ($/kWh)	Home	Load
Adrar	0.1	1	1	1	1,847	33,299	2.23	single	3.2kWh/day 628 W peak
	2	20	13	4	258,132	565,602	1.759	20	64kWh/day 13kW peak

7.4 Comparison with grid

In the present evaluation the distance of the proposed site (Adrar) is not identified however the breakeven grid extensions distance can be determined. The breakeven grid extension distance is the distance from the grid which makes the net present cost of extending the grid equal to the net present cost of the stand-alone system. Farther away from the grid, the stand-alone system is optimal. Nearer to the grid, grid extension is optimal. The figure 9(a) and 9(b) compare the costs of these two possibilities and show the breakeven grid extension is 3.71 km and 53.1 km respectively for single home and 20 homes, .

These off-grid options compete favorably with grid extension and so could be used in the current Algerian Energy Action Plan for the prevision of energy services to remote villages, schools, and rural health centers that are often located at distances outside the breakeven distances that have computed. These off-grid options can be used to reduce the number of villages that actually lack grid power supplies [24].

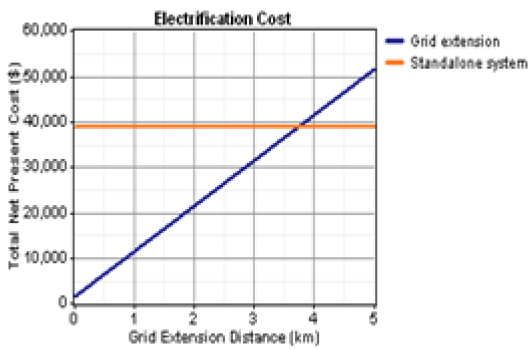


Fig.9 (a) The breakeven grid extension distance for single home

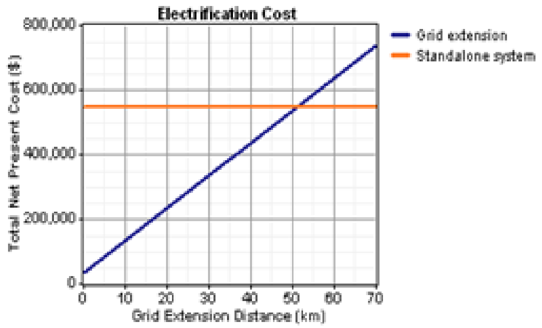


Fig.9 (b) The breakeven grid extension distance for 20 homes

7.5 Emissions

Table 4 (a) shows the emissions of the standard Diesel system (system 1), Wind/Battery /Diesel system (system 2) , PV/Battery/Diesel system (system 3) and the PV/Wind/Battery/ Diesel hybrid system (system 4) for single home and Table 4 (b) shows the emissions figures of the standard Diesel system (system 1), Wind/Battery/Diesel system (system 2) , PV/Battery/Diesel system (system 3) and the PV/Wind/Battery/Diesel hybrid system (system 4) for 20 homes. As the main greenhouse gas, the emission of carbon dioxide from system 1, system 2 and system 3 are 3,589, 335, 1,335 kg/yr while the optimized system exhausts only 268 kg/yr, which means a 91 % reduction for a single home and 95,850, 8532, 31,479 kg/yr while the optimized system exhausts only 7668 kg/yr, which means a 92% reduction for 20 homes.

Meanwhile, the sulphur dioxide and nitrogen oxide emissions of system 4 are equal to 0.539 and 5.91 kg/yr, respectively. While the emissions related to the first one of system 1, system 2 and system 3 are equal to 7.21, 0.624 and 2.68 kg/yr, respectively, while these emissions related to the second gas evaluate to 79, 7.39 and 29.4 kg/yr, for single homes, respectively. So, for 20 homes, the sulphur dioxide and nitrogen oxide emissions of system 4 are equal to 15.4 and 169 kg/yr, respectively. While the emissions related to the first one of system 1, system 2 and system 3 are equal to 192.5, 17.1, and 63.2 kg/yr, respectively, while these emissions related to the second gas evaluate to 2,012, 188 and 693 kg/yr, respectively.

Table IV(a) Annual emissions(kg/yr) of CO2, SO2 and NOx of each type of system for a single home

Pollutant emissions	system1	system2	system3	system 4
Carbon dioxide	3,589	335	1,335	268
Sulfur dioxide	7.21	0.674	2.68	0.539
Nitrogen oxides	79	7.39	29.4	5.91

Table IV (b) Annual emissions(kg/yr) of CO2, SO2 and NOx of each type of system for 20 homes

Pollutant emissions	system1	system2	system3	system 4
Carbon dioxide	48,473	8,532	31,479	7,668
Sulfur dioxide	97.3	17.1	63.2	15.4
Nitrogen oxides	1,068	188	693	169

8 CONCLUSION

This paper was devoted to simulating and sizing a PV/Wind/Diesel hybrid system with battery storage for rural electrification in Algeria. Different combinations of components size and count have been compared and the variations in resource availability and system costs influence on the installation and operation cost of different systems design explored. On the other hand, it could be summarized from analysis that it is more advantageous using the PV-Wind combination for 20-home rather than for single-home systems for which the per unit (kWh) cost of energy varies from 2.23 USD to 1.759 USD, which agree with the results obtained by Shamim Kaiser M and al [25].

The PV/Wind/Diesel hybrid system with battery storage combination for single-home systems has the lowest NPC which results in the shortest breakeven distance of 3.71 km.

In addition, the emissions of CO2, SO2 and NOx decrease to less than 90% of the standard diesel power system for single-home and for 20-home configurations.

Thus, the obtained results could be sufficient enough to enable decision making concerning the electrification of a remote area in a developing country [24], [26]. This constitutes a tool to evaluate and validate the high RE potential available in Algeria.

This study is still in its preliminary phase, and further investigations are needed in the following areas:

• Experimental analysis with a larger volume of data,

•Updating of the model considering transmission costs and other socio-environmental factors.

Finally, the overall system cost would be low if the turbine and battery costs decreases.

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